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Connector for shielded electric wire, and wire harness

Abstract

A connector for a shielded electric wire includes: a shield terminal connected to a braid on one end side thereof and grounding the braid by contact at a leaf spring on the other end side thereof; a first packing (sealing member) that seals between a housing and an object inserted into a through hole from an inner surface of the housing to an inner surface of the through hole; and a support member that is a resin ring exteriorly mounted on the shield terminal to be adjacent to the leaf spring and disposed inside the through hole in such a manner that an outer circumferential edge thereof is in contact with the inner surface of the through hole, thereby suppressing an offset of a central axis of the shield terminal with respect to a central axis of the through hole.

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Background/Summary

BACKGROUND

Technical Field

(1) The present invention relates to a connector for a shielded electric wire for connecting a shielded electric wire therethrough, and a wire harness with the connector provided at an end thereof.

Related Art

(2) Conventionally, in many cases, a connector for a shielded electric wire for connecting a shielded electric wire therethrough has a grounding structure for grounding a braid of the shielded electric wire to a metal panel constituting a connection target (for example, see JP 2019-9010 A). In the connector described in JP 2019-9010 A, a housing fixed to the metal panel of the connection target is made of metal, and the braid of the shielded electric wire is grounded to the metal panel via the metal housing.

(3) Here, in order to make it easy to manufacture housings, many housings are made of resins, and the housings made of resins may be adopted for certain connectors. It is usual that such a type of connector includes a shield terminal that grounds a braid to a metal panel of a connection target. As the shield terminal, there has been known a shield terminal formed in a tubular shape, with an end thereof being formed in a leaf spring shape, while being connected to the braid exposed from the shielded electric wire to cover the braid. In such a shield terminal, the leaf spring portion is in spring contact with an inner surface of a through hole for inserting a wire provided in the metal

panel of the connection target, thereby grounding the braid to the metal panel.

CITATION LIST

Patent Literature

(4) Patent Document 1: JP 2019-9010 A

SUMMARY

(5) Here, the shield terminal having the leaf spring structure described above is preferably in spring contact with the inner surface of the through hole with uniform contact pressure in a circumferential direction. To this end, it is preferable that the connector is fixed to the metal panel of the connection target in such a manner that a central axis of the tubular shield terminal does not deviate from a central axis of the through hole. However, the connector may be fixed to the metal panel of the connection target in a biased state or in an inclined state due to an influence of an error that occurs when a component is manufactured or the like. In such a case, an axis of the shield terminal may be offset with respect to the through hole, resulting in non-uniformity in contact pressure of the shield terminal in the circumferential direction with respect to the inner surface of the through hole, which may cause an increase ground resistance of the braid.

(6) Therefore, focusing on the above-described problem, an object of the present invention is to provide a connector capable of suppressing non-uniformity in contact pressure of a shield terminal in a circumferential direction with respect to an inner surface of a through hole. Another object of the present invention is to provide a wire harness with such a connector for a shielded electric wire provided at an end thereof.

(7) In order to solve the above-described problem, a connector for a shielded electric wire includes: a housing that is a tubular resin member configured to allow a shielded electric wire electromagnetically shielded by a braid to be inserted through the resin member, and is configured to be fixed to a connection target in such a manner that one end opening thereof communicates with a through hole provided in a metal panel constituting the connection target, and one end portion of the shielded electric wire extending from the one end opening passes through the through hole; a connector terminal provided at the one end portion of the shielded electric wire and configured to be connected to a terminal portion of the connection target; a shield terminal that is a metal tube configured to be externally mounted on the shielded electric wire, the shield terminal having one end side configured to be connected to the braid and another end side folded outward to form a leaf spring that generates a biasing force in a radially expanding direction, the leaf spring configured to come into contact with an inner surface of the through hole of the metal panel so as to press the inner surface of the through hole and to thus ground the braid to the metal panel; a sealing member formed of an elastic resin in a tubular shape to seal between the housing and an object inserted in the through hole from an inner surface of the housing to the inner surface of the through hole while avoiding a portion where the inner surface of the through hole and the leaf spring are in contact with each other; and a support member that is a resin ring externally mounted on the shield terminal to be adjacent to the leaf spring on a side opposite to the connector terminal, while being sandwiched between the leaf spring and the sealing member, an outer circumferential edge of the support member configured to be disposed inside the through hole in such a manner as to be in contact with the inner surface of the through hole, thereby suppressing an offset of a central axis of the shield terminal with respect to a central axis of the through hole.

(8) In addition, in order to solve the above-described problem, a wire harness includes: the above-described connector; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

(9) According to the above-described connector and wire harness, it is possible to suppress non-uniformity in contact pressure of the shield terminal in the circumferential direction with respect to the inner surface of the through hole.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view illustrating a connector for a shielded electric wire and a wire harness according to an embodiment;
 - (2) FIG. 2 is an exploded perspective view illustrating a configuration related to one wire in the connector and the wire harness illustrated in FIG. 1;
 - (3) FIG. 3 is a plan view of the connector illustrated in FIG. 1 as viewed from a direction of arrow V11 of FIG. 1;
 - (4) FIG. 4 is a cross-sectional view illustrating a cross section taken along line V12-V12 of FIG. 3 in the connector illustrated in FIG. 3; and
 - (5) FIG. 5 is an enlarged view of area A11 of FIG. 4 in the cross section illustrated in FIG. 4.
- ### DETAILED DESCRIPTION

- (6) Hereinafter, a connector for a shielded electric wire, and a wire harness will be described.
- (7) FIG. 1 is a perspective view illustrating a connector for a shielded electric wire, and a wire harness according to an embodiment. FIG. 2 is an exploded perspective view illustrating a configuration related to one wire in the connector and the wire harness illustrated in FIG. 1.
- (8) The wire harness 1 illustrated in FIG. 1 is routed in an automobile or the like, and a connector 1a for a shielded electric wire is provided therein at end portions of two shielded electric wires 1b.
- (9) The connector 1a is a two-terminal straight connector, and has a structure in which the two shielded electric wires 1b are inserted into a housing 11 formed of a resin in a double-tube type, and a connector terminal 12 is connected to an end of each shielded electric wire 1b. Here, the shielded electric wire 1b extending from the connector 1a and constituting the wire harness 1 is a wire electromagnetically shielded by a braid 1b-3 as illustrated in FIG. 2. The shielded electric wire 1b is formed in such a manner that a core wire 1b-1 is covered with a first sheath 1b-2, the first sheath 1b-2 is covered with a shielding braid 1b-3, and the braid 1b-3 is covered with a second sheath 1b-4. At the end of each shielded electric wire 1b, the connector terminal 12 is connected in a crimped manner to the core wire 1b-1 exposed after passing through the housing 11. In addition, the braid 1b-3 is exposed and folded outward at a position slightly away from the connector terminal 12 toward the housing 11. Then, the folded portion 1b-5 is connected to a shield terminal 13 for grounding the braid 1b-3 to a metal panel which is the ground on the connection target side.
- (10) In addition to the housing 11, the connector terminal 12, and the shield terminal 13 described above in outline, the connector 1a includes the following members. That is, the connector 1a includes a front holder 141, a rear holder 142, a support member 15, a first packing 161 (sealing member), a second packing 162, a shield ring 17, a protective material 18, and a shield sleeve 19. Hereinafter, these members will be described with reference to FIGS. 3 to 5.
- (11) FIG. 3 is a plan view of the connector illustrated in FIG. 1 as viewed from a direction of arrow V11 of FIG. 1, and FIG. 4 is a cross-sectional view illustrating a cross section taken along line V12-V12 of FIG. 3 in the connector illustrated in FIG. 3. FIG. 5 is an enlarged view of area A11 of FIG. 4 in the cross section illustrated in FIG. 4. FIGS. 4 and 5 illustrate cross sections of a connection target M of the connector 1a of the present embodiment and a metal panel M1 to which the connector 1a is fixed while constituting the connection target M.
- (12) As described above, the housing 11 is a double-tube resin member into which the shielded electric wires 1b are inserted. The housing 11 is fixed to the connection target M in such a manner that one end opening 111a thereof communicates with a through hole M11 provided in the metal panel M1 of the connection target M, and one end portion of the shielded electric wire 1b extending from the one end opening 111a passes through the through hole M11. In addition, the housing 11 includes a pair of tubular main body portions 111 and a flange 112 having a substantially rectangular shape as viewed in front thereof, the flange 112 being formed over the ends of the main

body portions **111** on the side where the connector terminals **12** are installed and spreading outward from the ends of the main body portions **111**. In addition, each of the shielded electric wires **1b** is inserted in each of the main body portions **111**. That is, the two shielded electric wires **1b** are inserted in the housing **11**. The flange **112** is a portion that is fixed to the metal panel **M1** of the connection target **M** in a surface-pressed manner. The one end openings **111a** of the pair of main body portions **111** are disposed at the flange **112**, and the connector terminal **12** is fixed to the end of the shielded electric wire **1b** extending from each of the one end openings **111a**. In addition, the flange **112** has a pair of attachment holes **112a** formed to penetrate therethrough in a plate thickness direction at diagonal positions as viewed in front thereof. Bolts (not illustrated) are inserted into the attachment holes **112a**, and the flange **112** is fixed to the metal panel **M1** by the bolts, and thereby, the connector **1a** is connected to the connection target **M**.

(13) The connector terminal **12** is a terminal metal fitting connected to a terminal portion inside the connection target **M**, and includes a connecting portion **121** having a flat plate shape and a crimped portion **122** provided at an end of the connecting portion **121** on the housing **11** side. The connecting portion **121** is a portion connected to the terminal portion by bolt fastening. The crimped portion **122** is a portion crimped while having the core wire **1b-1** in the exposed state.

(14) The shield terminal **13** is a metal tube externally mounted on the shielded electric wire **1b**, and a braid-connected portion **131** provided on one end side thereof is sheathed on and connected in a caulked manner to the folded portion **1b-5** of the braid **1b-3**, as will be described later. Then, a leaf spring **132** that generates a biasing force in a radially expanding direction **D11** is formed by folding the shield terminal **13** outward on the other end side thereof, and the leaf spring **132** is in contact with an inner surface of the through hole **M11** of the metal panel **M1** so as to press the inner surface of the through hole **M11** and to thus ground the braid **1b-3** to the metal panel **M1**.

(15) The front holder **141** is a tubular member made of a resin. When the side on which the shielded electric wire **1b** is connected to the connector terminal **12** is defined as a front side and the side opposite to the front side is defined as a rear side, the front holder **141** closes a space between the through hole **M11** of the metal panel **M1** and the shielded electric wire **1b** on the front side of the housing **11**. The front holder **141** includes a large-diameter portion **141a** that closes an outlet of the connection target **M** on an inner side thereof in the through hole **M11**, and a small-diameter portion **141b** that is inserted between the inner side of the leaf spring **132** of the shield terminal **13** and the shielded electric wire **1b**. The small-diameter portion **141b** supports pressure of the leaf spring **132** against the inner surface of the through hole **M11** from a central axis **G11** side of the through hole **M11**.

(16) The rear holder **142** is a tubular member made of a resin, and is a member that closes a space between a rear-side opening of the main body portion **111** of the housing **11** and the shielded electric wire **1b**.

(17) The support member **15** is a resin ring externally mounted on the shield terminal **13** to be adjacent to the leaf spring **132** on the side opposite to the connector terminal **12**. The support member **15** is disposed in such a manner that the leaf spring **132** is sandwiched between the support member **15** and the large-diameter portion **141a** of the front holder **141** in an axial direction **D12**, and the support member **15** is sandwiched between the leaf spring **132** and a first packing **161** to be described later. In addition, the support member **15** is disposed in such a manner that the shield terminal **13** is sandwiched between the support member **15** and the small-diameter portion **141b** of the front holder **141** in a radial direction **D13**. Also, the support member **15** is disposed inside the through hole **M11** in such a manner that an outer circumferential edge **15a** thereof is in contact with the inner surface of the through hole **M11**, thereby suppressing an offset of the central axis **G12** of the shield terminal **13** with respect to a central axis **G11** of the through hole **M11** of the metal panel **M1**.

(18) In the present embodiment, the support member **15** includes a tubular portion **151** inserted into a folded side of the leaf spring **132** of the shield terminal **13**, and a flange portion **152** protruding

from an opening edge of the tubular portion **151** on the side opposite to the leaf spring **132**. By having these portions, the support member **15** has an L-shaped cross section in the axial direction **D12**, and is in contact with the inner surface of the through hole **M11** at the outer circumferential edge **15a** of the flange portion **152**. The support member **15** is disposed in such a manner that the flange portion **152** faces an end **161a** of the first packing **161** facing the leaf spring **132**.

(19) In the present embodiment, the large-diameter portion **141a** of the front holder **141** is disposed in such a manner that the leaf spring **132** of the shield terminal **13** is sandwiched between the large-diameter portion **141a** and the support member **15**. Together with the outer circumferential edge **15a** of the support member **15**, an outer circumferential surface of the large-diameter portion **141a** of the front holder **141** is in contact with the inner surface of the through hole **M11**, thereby contributing to suppressing an offset of the central axis **G12** of the shield terminal **13** with respect to the central axis **G11** of the through hole **M11** of the metal panel **M1**.

(20) The first packing **161** is a sealing member formed of an elastic resin in a tubular shape to seal between the main body portion **111** and the object inserted in the through hole **M11** from an inner surface of the main body portion **111** of the housing **11** to the inner surface of the through hole **M11**. At this time, the first packing **161** performs sealing while avoiding a portion where the inner surface of the through hole **M11** and the leaf spring **132** are in contact with each other.

(21) The second packing **162** is a sealing member formed of an elastic resin in a tubular shape on the rear side of the main body portion **111** of the housing **11** to seal between the inner surface of the main body portion **111** and the shielded electric wire **1b** together with the first packing **161**. The second packing **162** performs sealing in a state where a shield sleeve **19** to be described later is sandwiched between the first packing **161** and the second packing **162**. On the rear side, the first packing **161** seals between the inner surface of the main body portion **111** and the shield sleeve **19**, and the second packing **162** seals between an inner surface of the shield sleeve **19** and the shielded electric wire **1b**.

(22) The shield ring **17** is a metal tubular member for connecting the braid-connected portion **131**, which is a tubular portion of the shield terminal **13** other than the leaf spring **132**, to the folded portion **1b-5** of the braid **1b-3** in a caulked manner. Here, a braid-connected portion **191** of the shield sleeve **19** is inserted into a portion caulked by the shield ring **17** between the folded portion **1b-5** of the braid **1b-3** and the second sheath **1b-4**. The shield ring **17** is externally mounted on a portion where the braid-connected portion **191** of the shield sleeve **19**, the folded portion **1b-5** of the braid **1b-3**, and the braid-connected portion **131** of the shield terminal **13** are stacked to caulk that portion by reducing its diameter.

(23) The protective material **18** is a tube member formed of a resin in a tubular shape and interposed between the first packing **161** and the portion caulked by the shield ring **17** to protect the first packing **161** from the caulked portion that is a metal stack.

(24) The shield sleeve **19** is a metal tubular member to be inserted into the rear side of the main body portion **111**. As described above, the shield sleeve **19** includes the braid-connected portion **191** which is connected to the folded portion **1b-5** of the braid **1b-3** in a caulked manner, together with the braid-connected portion **131** of the shield terminal **13**, by the shield ring **17**. The shield sleeve **19** includes a packing holding portion **192** having a larger diameter than the braid-connected portion **191** for the second packing **162** to be fitted therein, and a holder holding portion **193** having a larger diameter than the packing holding portion **192** to hold the rear holder **142**.

(25) According to the connector **1a** and the wire harness **1** described above, the support member **15** made of a resin ring is externally mounted on the shield terminal **13** adjacent to the leaf spring **132**. Also, the outer circumferential edge **15a** of the support member **15** is in contact with the inner surface of the through hole **M11** for inserting a wire in the metal panel **M1** of the connection target **M**, thereby suppressing an offset of the central axis **G12** of the shield terminal **13** with respect to the central axis **G11** of the through hole **M11**. According to the connector **1a** described above, by suppressing such an offset, it is possible to suppress non-uniformity of contact pressure of the

shield terminal **13** in a circumferential direction with respect to the inner surface of the through hole **M11**.

(26) Here, in the present embodiment, the support member **15** is a member having an L-shaped cross section in the axial direction **D12**, with a tubular portion **151** inserted into the folded side of the leaf spring **132** and a flange portion **152**. In addition, the support member **15** is in contact with the inner surface of the through hole **M11** at the outer circumferential edge **15a** of the flange portion **152**. According to this configuration, since the support member **15** is in contact with the inner surface of the through hole **M11** at the outer circumferential edge **15a** of the flange portion **152**, a contact area between the support member **15** and the inner surface of the through hole **M11** can be suppressed, thereby easily a clearance between the inner surface of the through hole **M** and the support member **15**.

(27) In the present embodiment, the support member **15** is disposed in such a manner that the flange portion **152** of the support member **15** faces the end **161a** of the first packing **161**. According to this configuration, since the contact between the end **161a** of the first packing **161** and the support member **15** in the axial direction **D12** is surface contact through the flange portion **152**, it is possible to easily restrict a movement of the support member **15** in the axial direction **D12**.

(28) In the present embodiment, the braid-connected portion **131** of the shield terminal **13** is connected to the folded portion **1b-5** of the braid **1b-3** in a caulked manner through the shield ring **17**. According to this configuration, the shield terminal **13** can be firmly connected in the caulked manner to the braid **1b-3** of the shielded electric wire **1b** electrically and mechanically.

(29) In addition, in the present embodiment, the protective material **18** formed of a resin in a tubular shape is interposed between the first packing **161** and the portion caulked by the shield ring **17** to protect the first packing **161** from the caulked portion. According to this configuration, since the first packing **161** is effectively protected from the caulked portion by the protective material **18**, the lifespan of the component can be extended.

(30) In the present embodiment, while the leaf spring **132** of the shield terminal **13** is sandwiched between the large-diameter portion **141a** of the front holder **141** and the support member **15**, the outer circumferential surface of the large-diameter portion **141a** is in contact with the inner surface of the through hole **M11** together with the outer circumferential edge **15a** of the support member **15**, thereby contributing to suppressing an offset of the axis. According to this configuration, the front holder **141** further suppresses the offset of the central axis **G12** of the shield terminal **13** with respect to the central axis **G11** of the through hole **M11**, making it possible to further suppress the non-uniformity of contact pressure of the shield terminal **13** in the circumferential direction with respect to the inner surface of the through hole **M11**.

(31) Note that the above-described embodiment merely shows typical forms of the connector and the wire harness. The connector and the wire harness are not limited the above-described embodiment, and various modifications can be made thereto.

(32) For example, the wire harness **1** routed in an automobile or the like is exemplified as an example of a wire harness in the above-described embodiment, but the wire harness is not limited thereto, and the specific location where the wire harness is routed is not limited.

(33) In the above-described embodiment, as an example of a connector, the connector **1a** that is a two-terminal straight connector is exemplified. However, the connector is not limited thereto, and the specific number of terminals and the like are not limited.

(34) In the above-described embodiment, as an example of a support member, the support member **15** contacting the inner surface of the through hole **M11** at the outer circumferential edge **15a** of the flange portion **152** is exemplified. However, the support member is not limited thereto, and the specific shape, the form of the contact with the inner surface of the through hole and the like are not limited as long as the support member can suppress an offset of the axis by contacting the inner surface of the through hole. According to the support member **15** contacting the inner surface of the

through hole **M11** at the outer circumferential edge **15a** of the flange portion **152**, a clearance between the inner surface of the through hole **M11** and the support member **15** can be easily reduced as described above.

(35) In the present embodiment, as an example of a support member, the support member **15** disposed in such a manner that the flange portion **152** faces the end **161a** of the first packing **161** is exemplified. However, the support member is not limited thereto, and the relative positional relationship of the support member with the sealing member around the wire and the like can be appropriately set. According to the support member **15** of which the flange portion **152** faces the end **161a** of the first packing **161**, a movement of the support member **15** in the axial direction **D12** can be easily restricted as described above.

(36) In addition, in the present embodiment, as an example of a shield terminal, the shield terminal **13** connected in a caulked manner to the folded portion **1b-5** of the braid **1b-3** through the shield ring **17** is exemplified. However, the shield terminal is not limited thereto, and any form can be adopted as a specific form of connection with the braid. According to the shield terminal **13** connected in a caulked manner to the folded portion **1b-5** of the braid **1b-3** through the shield ring **17**, the shield terminal **13** can be firmly connected to the braid **1b-3** as described above.

(37) In addition, in the present embodiment, as an example of a connector, the connector **1a** in which the protective material **18** is interposed between the first packing **161** and the caulked portion to protect the first packing **161** from the caulked portion is exemplified. However, the connector is not limited thereto, and no protective material may be provided between the caulked portion and the sealing member. As described above, by providing the protective material **18** between the caulked portion and the first packing **161**, the lifespan of the component can be extended.

(38) In the present embodiment, as an example of a connector, the connector **1a** in which a partial portion of the front holder **141** adjacent to the leaf spring **132** of the shield terminal **13** contributes to suppressing an offset of the axis is exemplified. However, the connector is not limited thereto, and the front holder may be away from the inner surface of the through hole in the metal panel, not contributing to suppressing an axial deviation. As described above, since the partial portion of the front holder **141** contributes to suppressing the offset of the axis, it is possible to further suppress the non-uniformity of contact pressure of the shield terminal **13** in the circumferential direction with respect to the inner surface of the through hole **M11**.

REFERENCE SIGNS LIST

(39) **1** Wire harness **1a** Connector for a shielded electric wire **1b** Shielded electric wire **1b-1** Core wire **1b-2** First sheath **1b-3** Braid **1b-4** Second sheath **1b-5** Folded portion **11** Housing **12** Connector terminal **13** Shield terminal **15** Support member **15a** Outer circumferential edge **17** Shield ring **18** Protective material **19** Shield sleeve **111** Main body portion **111a** One end opening **121** Connecting portion **122** Crimped portion **131, 191** Braid-connected portion **132** Leaf spring **141** Front holder **141a** Large-diameter portion **141b** Small-diameter portion **142** Rear holder **151** Cylindrical portion **152** Flange portion **161** First packing (sealing member) **161a** End **162** Second packing **192** Packing holding portion **193** Holder holding portion **D11** Radially expanding direction **D12** Axial direction **D13** Radial direction **G11, G12** Central axis **M** Connection target **M1** Metal panel **M11** Through hole

Claims

1. A connector for a shielded electric wire comprising: a housing that is a tubular resin member configured to allow a shielded electric wire electromagnetically shielded by a braid to be inserted through the resin member, and is configured to be fixed to a connection target in such a manner that one end opening thereof communicates with a through hole provided in a metal panel constituting the connection target, and one end portion of the shielded electric wire extending from the one end

opening passes through the through hole; a connector terminal provided at the one end portion of the shielded electric wire and configured to be connected to a terminal portion of the connection target; a shield terminal that is a metal tube configured to be externally mounted on the shielded electric wire, the shield terminal having one end side configured to be connected to the braid and another end side folded outward to form a leaf spring that generates a biasing force in a radially expanding direction, the leaf spring configured to come into contact with an inner surface of the through hole of the metal panel so as to press the inner surface of the through hole and to thus ground the braid to the metal panel; a sealing member formed of an elastic resin in a tubular shape to seal between the housing and an object inserted in the through hole from an inner surface of the housing to the inner surface of the through hole while avoiding a portion where the inner surface of the through hole and the leaf spring are in contact with each other; and a support member that is a resin ring externally mounted on the shield terminal to be adjacent to the leaf spring on a side opposite to the connector terminal, while being sandwiched between the leaf spring and the sealing member, an outer circumferential edge of the support member configured to be disposed inside the through hole in such a manner as to be in contact with the inner surface of the through hole, thereby suppressing an offset of a central axis of the shield terminal with respect to a central axis of the through hole.

2. The connector according to claim 1, wherein the support member is a member having an L-shaped cross section in an axial direction, with a tubular portion inserted into a folded side of the leaf spring on the another end side of the shield terminal and a flange portion protruding from an opening edge of the tubular portion on a side opposite to the leaf spring, and is configured to come into contact with the inner surface of the through hole at an outer circumferential edge of the flange portion.

3. The connector according to claim 2, wherein the support member is disposed in such a manner that the flange portion faces an end of the sealing member facing the leaf spring.

4. The connector according to claim 1, wherein the braid of the shielded electric wire is folded outward on a connection end side of the connector terminal, and a tubular portion other than the leaf spring of the shield terminal is connected in a caulked manner to the folded portion of the braid through a metal shield ring.

5. The connector according to claim 4, wherein the sealing member is externally mounted to the caulked portion of the shield terminal with the braid to cover the caulked portion, and a protective material formed of a resin in a tubular shape is interposed between the sealing member and the caulked portion to protect the sealing member from the caulked portion.

6. The connector according to claim 1, further comprising: a front holder that is configured to close a space between the through hole and the shielded electric wire on a front side of the housing, the front side being a side on which the shielded electric wire is connected to the connector terminal while a side opposite to the front side is defined as a rear side; and a rear holder that is configured to close a space between an opening on the rear side of the housing and the shielded electric wire, wherein the front holder is disposed in such a manner that the leaf spring of the shield terminal is sandwiched between a partial portion of the front holder and the support member, and an outer circumferential surface of the partial portion of the front holder is in contact with the inner surface of the through hole together with the outer circumferential edge of the support member, thereby suppressing the offset.

7. A wire harness comprising: the connector according to claim 1; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

8. A wire harness comprising: the connector according to claim 2; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

9. A wire harness comprising: the connector according to claim 3; and a shielded electric wire

electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

10. A wire harness comprising: the connector according to claim 4; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

11. A wire harness comprising: the connector according to claim 5; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.

12. A wire harness comprising: the connector according to claim 6; and a shielded electric wire electromagnetically shielded by a braid, the shielded electric wire extending from the housing with the connector terminal being provided at an end thereof.
